

# ECONOMICS

## Overview

[www.econ.iastate.edu](http://www.econ.iastate.edu) (<http://www.econ.iastate.edu>)

Economics teaches the ability to reason clearly and to address complex issues using tools and decision-making models from economics, mathematics, and statistics, as well as concepts from the biological, physical, and social sciences. The study of economics also helps students to: develop an understanding of the interactions of technology, human activity, and the environment; apply systematic approaches to making optimal choices; analyze quantitative information; and communicate concepts and findings to industry professionals, organizations, governments, and the general public.

Strong training in economic reasoning serves as a foundation for a variety of rewarding careers and for advanced study in a range of professional and academic fields. Economics majors are provided the tools of critical analysis and human relations skills that are essential for being informed citizens, finding and excelling in many career fields, and becoming lifelong learners.

### Student Learning Outcomes

Graduates from the Department of Economics at Iowa State University are expected to be able to:

1. Understand and apply concepts of scarcity, marginal analysis, opportunity cost and comparative advantage.
2. Critically analyze the efficiency and equity implications of market and non-market processes of allocation.
3. Evaluate policies, regulations, and business strategies using economic models, economic theories, and mathematical and statistical analyses.
4. Use quantitative and qualitative information to evaluate economic questions.
5. Understand the determinants of differences in economic performance over time and across space.

## Degree Requirements

### Economics Major, College of Liberal Arts and Sciences

The Economics major in the College of Liberal Arts and Sciences prepares students for advanced studies in Economics, Finance, and Analytics, for professional degrees such as medicine, law and business administration (MBA), and for careers in finance, business and economic research, management, insurance, brokerage, real estate, labor relations, international development, and government service. Candidates for the Bachelor of Science degree with a major in Economics must fulfill requirements established by the College of Liberal Arts and Sciences.

For details of undergraduate curricula in liberal arts and sciences, see College of Liberal Arts and Sciences (<https://catalog.iastate.edu/collegeofliberalartsandsciences/#lascollegerequirementstext>). A double major in Economics and Agricultural Business is permitted. A double major in Economics and Business Economics is not permitted. A major in Business Economics with a minor in Economics is not permitted. A minor in Economics cannot be combined with a major in Agricultural Business, and minors in both Agricultural Business and Economics are not permitted.

Students majoring in Economics are required to take the following courses within the General Education Area of Mathematics:

Choose one of the following pairs: 7-8

|                          |                                                         |
|--------------------------|---------------------------------------------------------|
| MATH 1600<br>& ECON 2070 | Survey of Calculus<br>and Applied Economic Optimization |
| MATH 1650<br>& ECON 2070 | Calculus I<br>and Applied Economic Optimization *       |
| MATH 1650<br>& MATH 1660 | Calculus I<br>and Calculus II                           |

Choose one of the following pairs: (STAT 2026 & STAT 3226) OR (STAT 3441 & STAT 3442): 6-8

|                          |                                                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| STAT 2026<br>& STAT 3226 | Introduction to Business Statistics<br>and Intermediate Business Statistics                                                         |
| STAT 3441<br>& STAT 3442 | Introduction to the Theory of Probability and<br>Statistics I<br>and Introduction to the Theory of Probability and<br>Statistics II |

Students planning to attend graduate programs in Economics or related fields are encouraged to take MATH 1650 Calculus I, MATH 1660 Calculus II, STAT 3441 Introduction to the Theory of Probability and Statistics I, and STAT 3442 Introduction to the Theory of Probability and Statistics II.

Students planning for careers in the finance industry or government sectors are encouraged to enroll in STAT 2026 Introduction to Business Statistics and STAT 3226 Intermediate Business Statistics.

Students must complete the following courses in economics:

|                                                                |                              |     |
|----------------------------------------------------------------|------------------------------|-----|
| ECON 1010                                                      | Principles of Microeconomics | 3   |
| ECON 1020                                                      | Principles of Macroeconomics | 3   |
| ECON 3010                                                      | Intermediate Microeconomics  | 3-4 |
| ECON 3020                                                      | Intermediate Macroeconomics  | 3   |
| ECON 3710                                                      | Introductory Econometrics    | 4   |
| Three credits of ECON 2300-2390, 3000-3890, 4000-4890 courses. |                              | 3   |

|                                               |   |
|-----------------------------------------------|---|
| Nine credits of 4000-4890 level ECON courses. | 9 |
| ECON 4920      Graduating Senior Survey       | R |

A minimum of 18 credits of economics coursework must be earned at Iowa State University. Economics majors must earn a minimum GPA of 2.0 across ECON 1010 Principles of Microeconomics, ECON 1020 Principles of Macroeconomics, ECON 3010 Intermediate Microeconomics, and ECON 3020 Intermediate Macroeconomics, with no grade in these lower than a C-.

Communication Proficiency Requirement: According to the university-wide Communication Proficiency Grade Requirement (<https://catalog.iastate.edu/academics/#communicationproficiencytext>), students must demonstrate their communication proficiency by earning a grade of C or better in ENGL 2500. In addition, the Economics major requires a grade of C or better in ENGL 3020 or ENGL 3140.

|           |                                                                         |   |
|-----------|-------------------------------------------------------------------------|---|
| ENGL 1500 | Critical Thinking and Communication                                     | 3 |
| ENGL 2500 | Written, Oral, Visual, and Electronic Composition                       | 3 |
|           | or ENGL 2500H Written, Oral, Visual, and Electronic Composition: Honors |   |
| ENGL 3020 | Business Communication                                                  | 3 |
|           | or ENGL 3140 Technical Communication                                    |   |

Students in all ISU majors must complete a three-credit course in both U.S. Cultures and Communities and in International Perspectives. The two courses may simultaneously meet the following graduation requirements: Social Science, Arts/Humanities, ECON 4000-4890, or Econ courses from approved list. Please discuss with your advisor.

LAS majors require a minimum of 120 credits, **including a minimum of 45 credits at the 3000/4000 level**. Three of the required 45 credits at the 3000+ level must be earned in a general education group outside the group of your major. Economics majors must meet or complete the LAS world language requirement and career proficiency requirement.

As majors in the College of Liberal Arts and Sciences, Economics students must meet College of Liberal Arts and Sciences (<https://catalog.iastate.edu/collegeofliberalartsandsciences/#lascollegerequirementstext>) and University-wide requirements (<https://catalog.iastate.edu/collegescurricula/>) for graduation in addition to those stated above for the major.

## Four Year Plan

### Bachelor of Science in Economics

#### Freshman

| Fall      | Credits Spring           | Credits |
|-----------|--------------------------|---------|
| ECON 1010 | 3 ECON 1020              | 3       |
| ENGL 1500 | 3 ECON 2070 <sup>a</sup> | 3       |

|                        |                          |           |
|------------------------|--------------------------|-----------|
| LIB 1600               | 1 STAT 2026 <sup>b</sup> | 3         |
| MATH 1600 <sup>a</sup> | 4 Arts and Humanities    | 3         |
| Arts and Humanities    | 3 Natural Science        | 3         |
|                        | <b>14</b>                | <b>15</b> |

#### Sophomore

| Fall                       | Credits Spring               | Credits   |
|----------------------------|------------------------------|-----------|
| ECON 3010                  | 4 ECON 3020                  | 3         |
| ENGL 2500                  | 3 STAT 3226 <sup>b</sup>     | 3         |
| LAS 2030                   | 1 Social Science             | 3         |
| Social Science             | 3 Natural Science            | 3         |
| Elective or World Language | 4 Elective or World Language | 4         |
|                            | <b>15</b>                    | <b>16</b> |

#### Junior

| Fall                                                        | Credits Spring                                                | Credits   |
|-------------------------------------------------------------|---------------------------------------------------------------|-----------|
| ECON 3710                                                   | 4 ECON 4000-4890                                              | 3         |
| ECON 2300-2390, 3000-3890, or 4000-4890                     | 3 Electives                                                   | 9         |
| Arts and Humanities                                         | 3 International Perspectives or U.S. Cultures and Communities | 3         |
| Elective                                                    | 2                                                             |           |
| International Perspectives or U.S. Cultures and Communities | 3                                                             |           |
|                                                             | <b>15</b>                                                     | <b>15</b> |

#### Senior

| Fall                | Credits Spring    | Credits   |
|---------------------|-------------------|-----------|
| ECON 4000-4890      | 3 ECON 4000-4890  | 3         |
| ENGL 3020 or 3140   | 3 ECON 4920       | R         |
| Arts and Humanities | 3 Natural Science | 2         |
| Electives           | 6 Electives       | 10        |
|                     | <b>15</b>         | <b>15</b> |

#### Total Credits: 120

Students in all ISU majors must complete a three-credit course in both U.S. Cultures and Communities and International Perspectives. The two courses may simultaneously meet the following graduation requirements: Social Science, Arts/Humanities, ECON 4000-4890, or Econ courses from approved list. Please discuss with our advisor.

LAS majors require a minimum of 120 credits, **including a minimum of 45 credits at the 3000/4000 level**. Three of the required 45 credits at the 3000+ level must be earned in a general education group outside the group of your major. Econ majors must meet or complete the LAS world language requirement. In addition, Econ majors must earn a

minimum of 18 credits from courses taught by the Department of Economics at ISU.

- a MATH 1650 (Calculus I) and MATH 1660 (Calculus II) may be substituted for MATH 1600 and ECON 2070.
- b STAT 3441 Introduction to the Theory of Probability and Statistics I and STAT 3442 Introduction to the Theory of Probability and Statistics II may be substituted for STAT 2260 and 3260.

## Minor

### Economics Minor, College of Liberal Arts and Sciences

For a minor in Economics, students complete a minimum of 15 credits. The minor must include at least 3 credits that are not used to meet any other Department, College, or University requirement and at least 9 credits at 2000+; and 6 of those at 3000+.

|                                                                        |                              |           |
|------------------------------------------------------------------------|------------------------------|-----------|
| ECON 1010                                                              | Principles of Microeconomics | 3         |
| ECON 1020                                                              | Principles of Macroeconomics | 3         |
| Six credits of Approved Economics Minor Courses List B at 3000+ level. |                              | 6         |
| Three credits of Approved Economics Minor Courses List A or B.         |                              | 3         |
| <b>Total Credits</b>                                                   |                              | <b>15</b> |

The Economics Undergraduate Minor is an LAS Minor. In addition to University policies governing minors (<https://catalog.iastate.edu/academics/#degreeplanningtext>), LAS minors require at least 9 credits in courses numbered 2000 and above; and 6 of those credits in courses numbered 3000 and above, with a grade of C or higher.

Approved Economics Dept. Minor Course List A: 2710, 3010, 3020, 3300, 3320, 3340, 3350, 3370, 3380X, 3530, 3620, 3640, 3800, 3820X, 4310, 4350, 4370, 4600, 4660, 4800

Approved Economics Dept. Minor Course List B: 3010, 3020, 3130, 3200, 3210, 3220X, 3440, 3530, 3550, 3710, 3760, 3800, 3820X, 3850, 4010, 4020, 4100X, 4160, 4180, 4310, 4500X, 4550, 4570, 4800

The Department of Economics has concurrent Bachelor of Science/ Master of Science (B.S./M.S.) programs designed especially for seniors who wish to pursue advanced studies. Concurrent BS/MS programs are economics (B.S.) with economics (M.S.) or agricultural economics (M.S.).

The concurrent B.S./M.S. program allows seniors to complete an M.S. degree within 12 months of B.S. completion.

For more information about our concurrent undergraduate and graduate programs in Economics, visit: <https://www.econ.iastate.edu/iowa-state-ms-econ-programs> (<https://www.econ.iastate.edu/iowa-state-ms-econ-programs/>).

## Graduate Programs

### Graduate Programs in Economics and Agricultural Economics

The Department of Economics offers work toward the degrees Master of Science and Doctor of Philosophy with majors in Economics and Agricultural Economics. The Department also offers minors to students with majors in other departments. For more information, visit our web site at [www.econ.iastate.edu](http://www.econ.iastate.edu) (<http://www.econ.iastate.edu>).

Students do not need to have an undergraduate major in Economics or Agricultural Economics to qualify for graduate work in the Department. However, students must have completed undergraduate coursework in macroeconomics, microeconomics, statistics, calculus, and matrix algebra. Some background in math courses emphasizing logic and proofs is preferred, particularly for the Ph.D.

Candidates for the degree Master of Science (thesis option) are required to complete satisfactorily 30 credits of acceptable graduate work, including preparation of a thesis.

Candidates for the degree Master of Science (non-thesis option) may fulfill requirements by satisfactorily completing 32 credits of acceptable graduate work, including preparation of a creative component.

Programs of study for the doctorate are organized by each student in consultation with the major professor and the student's committee. Subject to staffing constraints, the Department offers coursework to support the following fields of concentration: agricultural economics, applied econometrics, environmental and resource economics, financial economics, human resources, industrial organization, international economics, and macroeconomics. Students must complete advanced courses in microeconomic and macroeconomic theory, quantitative methods and econometrics, and two fields from the list above. Students are required to participate in workshops and demonstrate competence in theory by passing qualifying examinations.